Vadym Bartko

Python Backend Engineer / Software Engineer

Location: Vienna, Austria
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Visa & Work Permit: Valid work authorization in Austria. NO SPONSORSHIP REQUIRED
B2B: also available for contract work (via Gewerbe in Austria)
Languages: English (B2), German (A2)

Summary

- 15+ years in backend engineering and distributed systems (Python)
- Backend architecture: APIs, scalable microservices, document processing / data extraction pipelines
- Expertise in OCR and PDF processing
- Keywords: Python, Flask, Django, FastAPI, Celery, RabbitMQ, boto3, REST API, microservices, distributed systems, backend, async, PostgreSQL, SQL, MongoDB, Elasticsearch, AWS, S3, Lambda, SQS, SNS, EC2, Redshift, Textract, Azure, Document Intelligence, AI Services, Docker, Kubernetes, OCR, PDF, Ghostscript, PyMuPDF, ImageMagick, OpenAI, LLM, prompt engineering, data extraction, document processing, NLP, NLTK, OpenNLP, OpenCV, dlib, pandas, scikit-learn, Scala, GNU Radio, SDR, NumPy, SciPy, Embedded Linux, Git, GitHub, Jira, Linux, healthcare, fintech, crawling, scraping, object detection, face recognition, image classification, Cursor, Codex
- AI-assisted development (Cursor, Codex)

Skills

Backend: Python, Flask, Django, FastAPI, Celery, RabbitMQ, boto3, REST, microservices
Databases: PostgreSQL, SQL
Data processing / AI: OpenAI integration, AWS Textract / Azure Document Intelligence, image processing, NLP, some machine learning, text generation
PDF: Ghostscript, PyMuPDF, ImageMagick; PDF structure normalization and optimization
Cloud / ops: AWS (S3, Lambda, SQS, SNS, EC2), Azure, Docker, Kubernetes
Tools: Git, GitHub, Jira, Linux
Also: some OpenCV / NLP tools, applied ML / model deployment

Experience

P-Product - Python Backend Engineer (contractor)

Apr 2024 - Present · Vienna, Austria · healthcare

- Designed and implemented an OCR/ML pipeline for extracting blood-test results from medical documents with different layouts and languages.
- PDF preprocess, compression, OCR, LLM based extraction, marker names and values normalization.
- decrease processing from hours of manual work to 90 seconds
- Stack: Python, Flask, Django, Celery, RabbitMQ, PostgreSQL, Docker, OpenAI, Azure Document Intelligence / AI Services

Hardware Startup - counter-drone hardware (detection and jamming)

2023 - 2024

- Developed an SDR-based radio scanner and reactive jammer with automatic detection and suppression (incl. spoofing path).
- Hardware development, prototype, software-hardware integration, component selection, analysis of scanning and suppression technologies.

- Stack: GNU Radio, SDR, NumPy, SciPy, Embedded Linux, Orange Pi, UART

Zoral - Python Backend Engineer

2020 - 2023 · fintech

- Developed scalable backend services focused on PDF processing, AI integration, and AWS automation.
- Built and optimized processing pipelines for large batches of financial documents - multi-fold performance improvements, thousands of pages in 3-5 minutes.
- OCR of scanned pages, page cropping and filtering, compression and optimization of embedded images, overall file size reduction without loss of quality.
- Designed REST APIs and asynchronous workflows using Celery and RabbitMQ.
- Stack: Python, Flask, Celery, RabbitMQ, Ghostscript, PyMuPDF, ImageMagick, AWS (Lambda, S3, Textract)

Lohika - Machine Learning / Python Engineer

2015 - 2019

- **Bots / NLP:** customer-specific NLP bots; train and deploy ML apps to cloud.
  - Stack: Flask, Celery, Airflow, Kubernetes, Docker
- **Distributed training:** web APIs for third-party ML models; distributed training system; deployment and performance work.
  - Stack: Flask, Kubernetes, Docker
- **Media assets:** face detection/recognition, image tagging, NSFW/inappropriate-content detection.
  - Stack: OpenCV, Keras, TensorFlow, dlib, Docker
- **ITraffic:** empty parking-lot detection from webcam - datasets, models, API, deployment.
  - Stack: OpenCV, Caffe, Docker, RabbitMQ, AKAZE/ORB/BRISK, homography
- **Autonomous robocar:** control, navigation, mapping for a small 4W robot; SLAM experiments, hardware bring-up.
  - Stack: OpenCV, ROS, LIDAR, stereo camera
- **Gene variations:** classify pathogenic gene variants.
  - Stack: pandas, scikit-learn

Earlier career

2007 -2015 · Java / Python engineer, team lead

- **Crawling / scraping** - large-scale web crawling and data extraction.
  - Stack: Python, RabbitMQ, MySQL, PhantomJS, Redshift, S3, EC2, Elasticsearch, NLTK
- **Android / Java** - client apps and team lead.
  - Stack: Java, Android 2.x
- **Java backends** - product and outsourcing work.
  - Stack: Java, Spring, Hibernate, Maven, Gradle, Scala

Own projects

Khumbu (https://khumbu.im)

Ongoing

- Anki-style flashcards for learning words- extractd words from any photo or text, generate tranlation and samples
- OCR + LLM pipeline for vocabulary extraction and generation.
- Stack: Django, Docker, OpenAI, Celery

Image search engine

2018 - 2019 · alongside employment

- AI-powered image search engine with facial recognition, object detection, image classification, and metadata extraction.
- Search by plain text descriptions; filter by tags, metadata, faces, composition.
- Stack: Python, Flask, RabbitMQ, MongoDB, TensorFlow/Keras, Docker, Kubernetes, React

Nomad

- Distributed crawler.
- Stack: TitanDB, OrientDB, Gremlin

Firefly

- Web manager for scientific papers.
- Stack: GridFS, TitanDB, Elasticsearch

Denigma

open-source participation

- Ageing-research tooling: backend, knowledge extraction, some ops.
- Stack: CloudStack, Scala, NLTK, OpenNLP

BigPlanet

- Offline maps with multiple geo data sources.
- Stack: Android

Education

BS in Information Technology - Kharkiv National University of Radio Electronics (2008 - 2014)